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Digital Literacy, Labor Force Characteristics and the Degree of Adoption of Agricultural Socialized Services: Empirical Evidence from Rural China

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Abstract

Under the strategic goal of agricultural modernization, agricultural socialization services have become an important means of enhancing agricultural efficiency and guaranteeing food security. Based on microdata from 3811 farm households in seven provinces, this paper integrates labor force structural characteristics with digital literacy to construct a comprehensive analytical framework and empirically examines their effects on the degree of access to agricultural socialized services (DASS) through ordered logit model and moderated effects models. The results show that labor force characteristics significantly affect DASS, and the higher the degree of feminization, aging, and part-time employment, the higher the degree of access to services; digital literacy as a whole significantly improves DASS for farm households and shows heterogeneous moderating effects under different labor force characteristics. Therefore, this paper suggests formulating differentiated socialized service promotion strategies, deepening the digitalization of agricultural services, strengthening the digital technology training of rural laborers in various ways, enhancing DASS, effectively improving the efficiency of agricultural production, and supporting the dual goals of food security and rural revitalization.

Keywords: digital literacy; labor force characteristics; agricultural social services; ordered logit model

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1. Introduction

Agricultural production is the "greatest of nations," and guaranteeing the effective supply of important agricultural products and promoting the sustainable development of agriculture have become the core issues of national governance. With the promotion of key agricultural strategies such as "storing grain in technology and storing grain on the land," China's agricultural development is entering a new phase centered on productivity improvement. However, the structural change in the agricultural labor force has brought serious challenges to the process of agricultural modernization; the rural population continues to flow out, the agricultural labor force shows structural characteristics such as aging, feminization, and part-time work, and the problems of labor force rigidity constraints and weaknesses are becoming more and more prominent, and the questions of "who is going to farm" and "how to farm" have become the constraints to agricultural modernization. In this context, agricultural socialized services act as a link between the various aspects of agricultural production to achieve the "people retired technology into" an important institutional arrangement and gradually become an

important path to alleviate the labor rigidity constraints, enhance the efficiency of agricultural production, and promote the modernization of agriculture [1,2]. Agricultural socialized services through the provision of plowing, planting, prevention, harvesting, marketing, and other specialized, large-scale support services reduce the technical threshold of small farmers and production costs and effectively improve the efficiency of agricultural production and risk-resistant capacity [3–5], the rise of which cracked the "who will plant the land" and "how to plant the land" questions. Its rise has provided new answers to the problem of "who will farm the land" and "how to farm the land." Central Document No. 1, as the first document issued by the Chinese government each year, focuses on issues related to agriculture, has also repeatedly emphasized the need to improve the socialized service platform, improve the standard system, strengthen its fundamental role in the modern agricultural system, and build a unified, open, competitive, and orderly market for modern agricultural services. However, in practice, agricultural socialized services are still facing the reality of imbalanced regional development, insufficient willingness of farmers to participate, and a poor match between service supply and demand, which urgently requires an in-depth understanding of the influencing factors behind their services at the level of farmers [6]. While prior studies have tended to focus either on labor force structure or on digitalization in agriculture, few have integrated these two dimensions into a unified framework. This paper advances the literature by simultaneously examining labor force characteristics and digital literacy as joint determinants of farmers' access to agricultural socialized services, thereby providing a more comprehensive understanding of the mechanisms service adoption.

2. Literature Review

The factors influencing farmers' access to agricultural socialized services are complex and diverse, and existing research has focused on farmers' characteristics, household resource endowment, and land management characteristics [3,7]. From the point of view of individual and family characteristics of farm households, factors such as gender, age, education level, the number of family laborers, and the size of arable land not only independently affect their agricultural socialized service adoption behavior but are also associated with non-farm employment and family income generation, which jointly play a role in the perception and adoption process of agricultural socialized services by farm households [8–10]. In addition, as the labor force structure continues to evolve, the physical and cognitive deterioration of aging, the traditional gender division of labor of feminization, and the lack of time and energy of part-time work have all weakened the agricultural production capacity of farm households to a certain extent and increased their dependence on agricultural socialization services [11]. Although the rigid constraints of the agricultural labor force limit subsistence agricultural production, it objectively promotes the tendency of farm households to outsource production links and accept specialized services. Accordingly, this paper selects three indicators of labor force characteristics, namely aging, feminization, and part-time employment, to fit the real trend of agricultural labor force structural transformation. From the point of view of changes in labor force characteristics, the aging, feminization, and part-time employment of the agricultural labor force not only affect the efficiency of agricultural production and the production and management decision-making of farmers but also indirectly promote farmers to increase the demand for agricultural socialized services and reshape their perception of socialized services [12–15]. At the land level, land scale is an important factor influencing the decision-making of agricultural socialized service adoption behavior, and the fragmentation of arable land restricts the process of agricultural mechanization [16], while the large-scale operation of land strongly promotes the acquisition of agricultural socialized services [3]. Recent international literature increasingly underscores the importance of digital empowerment

in agricultural adoption literature. For example, sustainability studies show that digital technologies can significantly increase growers' economic benefits by reducing costs and raising yield and quality—illustrating digital tools' universal relevance [17]. In low- and middle-income countries, digital tools and data-driven services are becoming critical components of sustainable food systems, helping smallholders overcome access barriers [18]. Moreover, peer-to-peer digital extension platforms—such as low-cost SMS-based systems—are emerging in global South contexts, improving inclusion and adoption among smallholders [19]. In addition, region-specific reviews from South Africa highlight how infrastructure gaps, digital illiteracy, and gender inequality hinder digital adoption among smallholder farmers, stressing that digital divide challenges must be addressed for equitable digital transformation [20]. On this basis, some studies have proposed the concept of "digital literacy" to measure the actual ability of farmers to use digital technology to solve agricultural problems, emphasizing its positive role in promoting agricultural input behavior, access to agricultural service information, and improving market participation capacity [21–23].

The above research has laid a solid foundation for studying agricultural socialized services. However, there remains a lack of research on the impact of labor force characteristics on the extent to which agricultural socialized services are accessed. Empirical studies on the influence of farmers' digital literacy on their behavior in accessing agricultural socialized services are also relatively scarce. In the process of agricultural modernization, differences in labor force structure and quality will inevitably manifest as varying DASS, thereby exerting a profound impact on agricultural production efficiency and agricultural development. This prompts us to consider: How do changes in labor force structure influence farmers' access to agricultural socialized services? Does digital literacy significantly moderate this influence mechanism? Does this moderation mechanism exhibit regional differences? These questions warrant further exploration.

3. Theoretical Analysis and Hypotheses

3.1. Influence Mechanisms of Labor Force Characteristics on Access to Agricultural Socialized Services

As the agricultural labor force continues to transfer to non-agricultural fields, agricultural production shows obvious characteristics of an aging labor force, feminization, and part-time employment, and the rigid constraints of the agricultural labor force affect the access of farm households to agricultural socialized services through multiple pathways. An aging labor force drives the growth of demand for agricultural socialized services in two ways: on the one hand, an aging labor force is unable to independently complete high-intensity, technology-intensive farming activities due to physical decline and insufficient ability to adapt to new technologies, so it turns to rely on professional services to make up for its lack of labor efficiency; on the other hand, aging families are more inclined to retain the right to operate the land to protect their basic livelihoods, but their production capacity constrains them, and they need to outsource services to achieve the goal [11,24]. On the other hand, aging families prefer to retain their land management rights to secure their basic livelihoods but are limited by their production capacity and need to outsource services to realize effective land use. Secondly, under the feminized labor structure, under the dual pressure of family caregiving and non-farm employment, women's time and energy investment in agricultural production is limited, coupled with the fact that part of the agricultural production process may have certain physiological or skill thresholds for women, which has prompted female-dominated households to prefer replacing traditional agricultural production through socialized services [25,26]. In addition, non-farm employment also leads to a shortage of agricultural labor, and part-time farmers seek a balance between

non-farm employment and agricultural production and need to use socialized services to maintain agricultural production, while the increase in non-farm income improves their ability to pay, forming a positive cycle of "income increase—service purchase." Based on the above analysis, this study proposes the following hypotheses:

H1. *The aging, feminization, and part-time employment of the agricultural labor force promote the acquisition of agricultural socialized services.*

3.2. Influence Mechanism of Digital Literacy on DASS

Digital literacy is the ability to access, understand, and integrate digital information for a productive life in the era of the digital economy [27]. With the popularization of digital technology in the agricultural field and the improvement of farmers' digital skills, digital literacy has become a crucial factor influencing farmers' adoption decisions of socialized services. According to the Technology Acceptance Model (TAM) [28], digital literacy can have an impact on farmers' access to agricultural socialized services through the perceived ease of use and perceived usefulness of the services: on the one hand, farmers with higher digital literacy can skillfully use various digital platforms and tools, effectively break through the traditional spatial and temporal barriers of information transfer, and search, filter, and compare information from different socialized service providers more conveniently and at lower costs, reducing the cost of information transfer. The information of different socialized service providers reduces the decision-making risk and search cost caused by information asymmetry and uncertainty, which is conducive to a clearer understanding of the potential value of socialized services and enhances the perceived ease of use of the services, thus making it easier to complete the selection of the services and access to them; on the other hand, farmers with higher levels of digital literacy can grasp the information about the dynamics of the market for agricultural products, the price trend, the preference of consumers, and the reputation evaluation of service providers in a more timely and comprehensive manner. On the other hand, farmers with higher levels of digital literacy can more timely and comprehensively grasp the market dynamics of agricultural products, price trends, consumer preferences, as well as the credibility of the service provider's evaluation and other information advantages, which can help to more accurately predict the adoption of a particular socialized service may bring about the economic benefits and market competitiveness to improve, alleviate the asymmetry of information and enhance the market expectations, which enhances the perceived usefulness of the service, thereby strengthening the positive expectations for the service's effectiveness, and thus enhancing DASS. Thus, the hypothesis is proposed:

H2. *The higher the digital literacy, the more farmers tend to access agricultural socialization services.*

3.3. Moderating Mechanism of Digital Literacy in the Influence of Labor Force Characteristics on DASS

Digital literacy lowers the threshold for older farmers to access service information and compare service quality and price, effectively overcomes information asymmetry or operational inconvenience due to the age factor, and further stimulates their latent service demand [29]. At the same time, part-time farmers can make use of tools such as digital management platforms and precision agriculture applications to more efficiently plan agricultural affairs, dock service resources, and supervise the service process, making the purchase of services more time-efficient and output-effective and enhancing their motivation to optimize agricultural production through service outsourcing. However,

through online learning platforms and digital agricultural extension, women farmers can conveniently acquire agricultural knowledge and skills and enhance their ability to independently complete some production processes, which may partially replace their reliance on socialized services for certain aspects of production or post-production and may inhibit or adjust their direct demand for traditional socialized services to a certain extent [30]. Therefore, digital literacy, as a kind of digital information utilization ability, can play a significant moderating effect between labor force characteristics and agricultural socialized service access, and the direction and intensity of this moderating effect will show differential performance according to labor force characteristics; accordingly, the hypothesis is proposed:

H3. *Digital literacy has a significant moderating effect on the relationship between labor force characteristics and DASS, but the direction of the moderating effect varies depending on the labor force characteristics.*

Based on H1–H3, this study develops a conceptual framework illustrating labor force characteristics' impact mechanisms on DASS (see Figure 1).

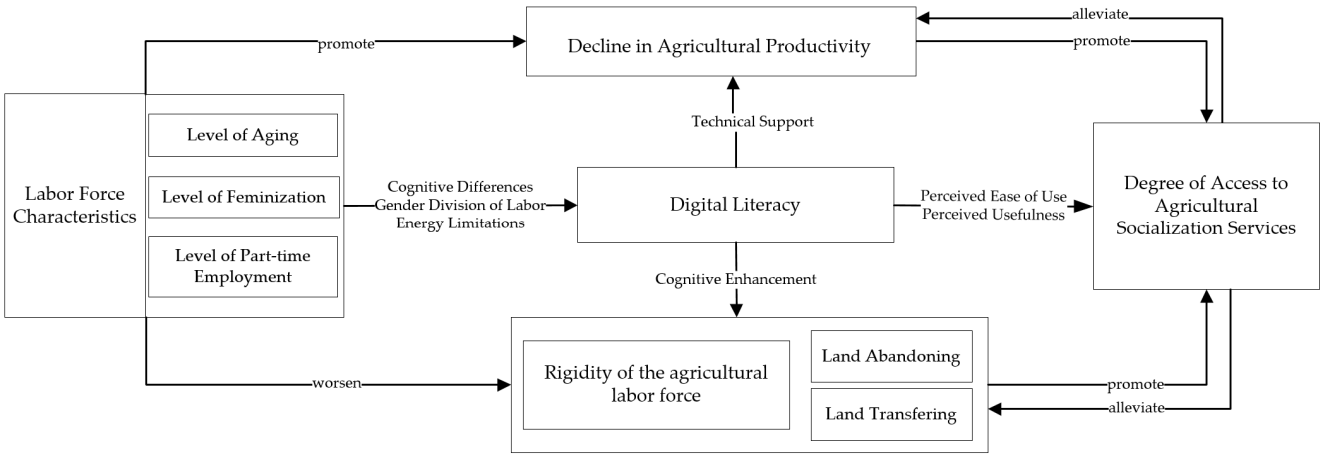


Figure 1. Mechanism of labor force characteristics' impact on DASS.

4. Materials and Methods

4.1. Data Sources

The data used in this paper come from the China Rural Microeconomic Survey conducted by the Institute of Agricultural Economics and Development of the Chinese Academy of Agricultural Sciences (CAAS) in 2021, of which the survey of farmers involves the basic information of farming households, agricultural production, land resource utilization, household income and expenditure, consumption, and the access to agricultural socialized services, etc., and the survey of villages includes the basic situation of the villages, the state of economy and development, and the governance and social development of villages. The sample area covers seven provinces, including Jilin, Sichuan, Hebei, Hunan, Xinjiang, Fujian and Zhejiang, covering the three major grain-producing areas and three types of terrain: plains, hills and mountains, and involves 25 cities and counties, each of which selects three townships and nine administrative villages according to the good, medium and poor economic conditions, and randomly samples 20 households from each administrative village, and then removes the samples that have obvious outliers and missing information after data cleaning. After data cleaning to exclude samples with obvious outliers and serious information loss, we finally obtained 3811 samples from 199 villages in 7 provinces (Table 1).

Table 1. Sample distribution.

Province	Fujian	Hebei	Hunan	Jilin	Sichuan	Xinjiang	Zhejiang	Total
Number of Villages	27	21	27	27	27	43	27	199
Number of Households	532	421	530	537	540	714	537	3811
Sample Proportion	13.96	11.05	13.91	14.09	14.17	18.74	14.09	100

4.2. Variables Selection

4.2.1. Dependent Variable: *DASS*

There is a diversity of methods to measure DASS in the academic community, some studies from the perspective of capital input, using the cost of socialized services adopted by farmers, or the proportion of the cost of total inputs to agricultural production, as a quantitative indicator of DASS [31,32]; other studies examine whether farmers adopt socialized services in agricultural [33–35]. In addition, some scholars try to use the number of socialized service items purchased by farmers as the basis for measurement [36]; however, this method of directly adding up the number of items may not fully consider the importance of the differences between different service segments, and it is difficult to accurately reflect the actual DASS. To more comprehensively reflect the degree of farmers' reliance on different agricultural socialization services, this paper draws on the research method of Tang et al. [37], taking into account the fact that any agricultural production link is closely related to crop yields, and that all links are equally important, DASS by farmers is subdivided into four tiers: access to fewer than three services is assigned the value of 0, access to 3–5 services is assigned the value of 1, access to 6–10 services is assigned the value of 2, and access to more than 11 services is assigned as 3.

4.2.2. Explanatory Variables: *Labor Force Characteristics*

To comprehensively examine the impact of the changing structural characteristics of the labor force on DASS, this paper defines labor force characteristics from the three dimensions of labor force aging, feminization, and part-time employment, taking into account the quantitative characteristics of the labor force and the attributes of the quality of the labor force. Refer to existing study [38,39], the age of aging is defined as over 60 years old; aging and feminization are measured as the ratio of the number of aged labor force in the household and the number of female labor force in the household to the total labor force in the household, respectively, while part-time employment is measured by the ratio of wage income to the total income of the household.

4.2.3. Moderating Variable: *Digital Literacy*

In this paper, the digital literacy of farmers is selected as a moderating variable, and concerning Eshet's study [40], digital literacy is defined as the ability to utilize internet information for agricultural production and life and is measured by the three indicators of digital farming, digital business information, and digital fitness (Table 2). The entropy value method was utilized to determine the weights corresponding to the three indicators of digital literacy based on the research data to derive the digital literacy values [41].

Table 2. Digital literacy indicators system.

Indicator	Meaning	Mean	Standard Deviation	Weights
Digital Farming	Whether the internet is used to access information on agricultural production [1 = YES, 0 = NO]	0.361	0.480	0.333
Digital Newsletter	Whether the internet is used to obtain information on agricultural markets [1 = YES, 0 = NO]	0.333	0.471	0.333
Digital Health	Whether the internet is used to access food nutrition and health information [1 = YES, 0 = NO]	0.344	0.475	0.333

4.2.4. Control Variables

Refer to the studies of Tong et al. and Yang et al. [42,43], a total of 14 control variables are selected from three categories: individual characteristics, household characteristics, and village characteristics. Among them, individual characteristics include average years of education in the family, average health status in the family, experience of serving as village cadres, and whether they have received agricultural and non-agricultural training; family characteristics include sown area, number of plots, possession of self-purchased agricultural machinery and equipment, total income from agricultural production services, and participation in professional cooperatives; and village characteristics include the topography of the village, geographic location, presence of farmers' professional cooperatives, the number of agricultural extension services and trainings carried out in this year, and the grain production area to which it belongs. Training in the current year and the grain production area to which it belongs.

4.3. Method Section

This paper focuses on the impact of digital literacy and labor force characteristics on DASS of farmers. With the help of models such as Probit and Logit [11,42], scholars have dissected the influence of the above factors on the behavior and DASS which provides theoretical and empirical support for understanding the decision-making of the agricultural socialized service acquisition behavior of farmers. Considering that the dependent variables of DASS are ordered discrete variables, we choose the ordered logit regression model for estimation, and the specific model settings are as follows:

$$logit(P(Y \leq j)) = \alpha_j + \beta_1 female + \beta_2 old + \beta_3 div + \Sigma\{i = 1\}\hat{}\{15\} Y_i C_i + \delta_i, j = 0, 1, 2 \quad (1)$$

where $logit(P \leq j) = ln\,[(P \leq j)/(1 - (P \leq j))]$, α_j is the intercept term; different j corresponds to different intercepts, which is used to classify the dependent variable Y; β_1, β_2, and β_3 are the regression coefficients of the explanatory variables female, old and div, respectively; C_i are a set of control variables, including individual characteristics of farmers, family business characteristics, and village characteristics; γ_i are the coefficients of each control variables, and δ_i are the random error terms.

To explore the influence of digital literacy in labor force characteristics on DASS, we add the cross terms of labor force characteristics (*female*, *old* and *div*) and digital literacy (*internet*) in the above model and construct the moderating effect model as follows:

$$logit(P(Y \leq k)) = \alpha_k + \beta_1 female + \beta_2 old + \beta_3 div + \beta_4 internet + \beta_5 fi + \beta_6 oi + \beta_7 di + \Sigma\{i = 1\}\hat{}\{15\} Y_i C_i + \eta_i, k = 0, 1, 2 \quad (2)$$

where β_i is the parameter to be estimated for the corresponding variable, and $female \times dl$, $old \times dl$, and $div \times dl$ are the cross terms of labor force characteristics and digital literacy, and η_i is the regression coefficient of the random error term.

5. Results

In this paper, we use Stata18.0 software to quantitatively estimate the effects of digital literacy and labor force characteristics on DASS, and before the regression estimation, all the variables are tested for multicollinearity, and the mean value of Variance Inflation Factor (VIF) is 1.33 which is less than 10, and based on the criterion of judging the multicollinearity [44]. The model does not have a serious multicollinearity problem.

5.1. Descriptive Statistics and Characteristic Facts

As shown in Table 3, the overall level of access to agricultural socialized services among the sample households is relatively low, with a relatively concentrated distribution and a

mean value of 1.163. The proportion of the female labor force in the sample households is relatively stable, but there are significant differences in the degree of aging. The standard deviation of the degree of diversification exceeds 30%. The mean value of digital literacy is 0.346, with a standard deviation greater than the mean value, indicating that farmers' digital skills exhibit a polarized distribution. At the individual and household levels, the average household health level is between healthy and sickly, the average education level is junior high school, and the enthusiasm of farmers to participate in skills training is medium to high, which is in line with the typical characteristics of the rural population, and the centralized distribution of the experience of serving as village cadres is in line with the characteristics of the rural grass-roots governance structure. The dispersion coefficients of the total sown area of households and the number of plots are as high as 4.13 and 11.41, respectively, indicating that the scale of land operation is highly unbalanced, with small and large farmers coexisting; fragmentation is still prominent; and the rates of self-purchase of agricultural machinery and participation in cooperatives are at a relatively low level, with significant differences in the returns from agricultural production. At the village level, the study area is mainly hilly terrain, with generally well-developed transportation and high cooperative coverage but low agricultural machinery promotion services and training frequency (only 2.05 times per year on average), indicating that the intensity of village-level agricultural service promotion is weak.

Table 3. Variables' definitions and descriptive statistics (N = 3811).

Variable	Meaning	Mean	SD
DASS	Total number of items: [0 = (X < 3); 1 = (3 ≤ X ≤ 5); 2 = (6 ≤ X ≤ 10); 3 = (11 ≤ X)]	1.163	0.855
Female	Share of female labor force in the household labor force	0.264	0.211
Old	Share of older labor force in household labor force	0.314	0.346
Div	Share of wage income in total household income	0.468	0.339
Digital literacy	The results were calculated using the entropy method [1]	0.346	0.441
Individual characteristic			
Healthiness	Average health of the household: [1 = healthy; 2 = frail; 3 = chronic; 4 = major illness; 5 = disability]	1.377	0.877
Edu	Average years of schooling of the labor force(year)	7.589	3.07
Village cadre experience	Whether any member of the household is a village cadre: [1 = YES, 0 = NO]	0.863	0.344
Training participation	Has anyone in the household participated in agricultural/non-agricultural training: [1 = YES, 0 = NO]	0.548	1.049
Characteristics of household business			
Sown area	Total sown area (mu [2])	32.477	134.241
Number of plots	Number of plots(number)	6.487	74.025
Farm machinery	Whether the household has purchased its agricultural machinery: [1 = YES, 0 = NO]	0.219	0.414
Agricultural income	Gross income from agricultural production services (CNY 10,000)	5.231	22.9
Specialized associations Participation	Household participation in professional associations or professional cooperatives: [1 = YES, 0 = NO]	0.108	0.31
Village Characteristics			
Village terrain	Topography of the village [3]: [1 = plains; 2 = hills; 3 = mountains]	1.837	0.808
Geographic location	Distance of this VDC from the county government (km)	25.031	18.272
Promotion of agricultural machinery	Number of agricultural extension services and training conducted in the village(number)	2.053	6.269
Village cooperative	Whether there is a cooperative in the village: [1 = YES, 0 = NO]	0.724	0.447
Grain-producing region	Grain-producing areas [4]: [1 = main producing areas; 2 = balanced areas; 3 = main consumption areas]	1.746	0.867

[1] The entropy method is used to determine the corresponding weights of the three indicators of digital literacy, which results in the value of digital literacy. [2] 1 mu = 667 m^2 or 0.067 ha. [3] Plains are areas with an altitude of less than 200 m, hills are between 200 and 500 m, and mountains are above 500 m. [4] Main producing areas are the suppliers of grain to the national market, and the balanced areas are regions where grain production is roughly equal to local consumption demand, while the main consumption areas refer to regions where local grain production is insufficient to meet consumption demand.

Further from regional heterogeneity (Table 4), it is found that there are obvious geographical differences in DASS, labor force characteristics, and the level of digital literacy.

The mean value of DASS in Xinjiang (1.87) is much higher than the sample mean, indicating a significant synergy between its large-scale agricultural production operations and policy support. On the contrary, Zhejiang and Fujian Provinces, as the major grain-consumption regions, have relatively low service levels, suggesting that market orientation and channel construction in the major grain-consumption regions may have had a complex impact on the access to agricultural socialization services. In terms of digital literacy, Hunan, Xinjiang and Fujian are "moderately high", while Zhejiang scores lower but still maintains a service level above the medium level, suggesting that the role of digitalization may be constrained by multiple factors such as the characteristics of agricultural production, the structure of the labor force and market orientation in different regions.

Table 4. Characteristic facts of access to agricultural socialized services, labor force characteristics, and digital literacy.

Variable	Mean	Major Grain-Producing Regions				Balanced Areas	Major Grain-Consumption Regions	
		Hebei	Hunan	Jilin	Sichuan	Xinjiang	Zhejiang	Fujian
DASS	1.16	1.47	1.27	1.34	0.49	1.87	0.89	0.65
Female	0.26	0.24	0.26	0.26	0.23	0.31	0.24	0.28
Old	0.31	0.31	0.29	0.33	0.39	0.20	0.41	0.30
Div	0.47	0.47	0.58	0.45	0.43	0.44	0.41	0.51
Digital literacy	0.35	0.36	0.40	0.26	0.38	0.42	0.20	0.37
Samples	3811	421	530	537	540	714	537	532

5.2. Analysis of Benchmark Regression Results

From Table 5, it can be seen that labor force characteristics have a significant positive effect on DASS. Specifically, the three variables of feminization, aging and part-time employment are significantly positively correlated at the statistical levels of 5%, 1% and 1%, respectively, indicating that as the degree of feminization, age structure and occupational differentiation of the agricultural labor force increases, the dependence of farmers on external specialized services increases significantly, and this result is robustly verified in Models 1 to 4, thus, Hypothesis 1 is validly argued.

With the gradual introduction of various control variables (Models 2 to 4), the individual characteristics, the health level of the head of household and the education level have a positive impact on the service acquisition, and the experience of serving as a village cadre significantly improves the level of service adoption through the advantages of policy information and interpersonal network resources; the positive effect of the training and participation behaviors verifies the logic of "skill adaptation promotes the choice of service". The positive effect of training participation behavior verifies the logic of "skill adaptation promotes service selection". At the household level, sowing area, farm income and purchasing farm machinery all have a positive and significant effect on the acquisition of agricultural socialized services, reflecting the high demand for socialized services by large-scale operators; while the number of plots inhibits the acquisition of services, revealing the dilemma of "fragmented operation increases the cost of service coordination". In terms of village characteristics, it is difficult to realize the effective promotion of mechanized services in mountainous areas or areas with large topographic relief, while geographic location has a significant positive effect, suggesting that remote areas tend to rely on services to supplement the labor gap due to the exodus of laborers. However, the results of "individual participation positive, overall coverage negative" at the level of cooperatives, which may stem from the inefficiency of some cooperatives or the existence of the problem of "existing in name only", or the formation of substitution with the farmers

to purchase their farm machinery, indicate the need for further research on the quality of cooperative services to clarify the direction of policy optimization. Further research on the quality of cooperative services is needed to clarify the direction of policy optimization. Finally, the significant negative effect of the type of grain production area indicates that the main grain production area has significantly improved the accessibility and utilization of services for farmers by virtue of its centralized and contiguous area, policy support, and service network maturity. In contrast, the main marketing areas are dominated by market mechanisms and lack strong policy traction, and their agricultural service systems are still in the exploratory stage.

Table 5. Impact of labor force characteristics on DASS.

Variable	Model 1	Model 2	Model 3	Model 4
Female	0.435 **	0.436 **	0.502 **	0.472 **
	(0.197)	(0.197)	(0.199)	(0.201)
Old	1.091 ***	0.784 ***	0.584 ***	0.561 ***
	(0.120)	(0.123)	(0.125)	(0.127)
Div	0.773 ***	0.709 ***	0.381 ***	0.173 **
	(0.079)	(0.080)	(0.085)	(0.088)
Healthiness		−0.131 ***	−0.099 ***	−0.084 **
		(0.036)	(0.036)	(0.037)
Education		0.054 ***	0.049 ***	0.028 **
		(0.011)	(0.011)	(0.011)
Village cadre experience		0.289 ***	0.375 ***	0.314 ***
		(0.091)	(0.092)	(0.093)
Training participation		0.359 ***	0.306 ***	0.291 ***
		(0.030)	(0.031)	(0.031)
Sown area			0.004 ***	0.003 ***
			(0.001)	(0.001)
Number of plots			−0.005 ***	−0.003 ***
			(0.001)	(0.001)
Farm machinery			0.555 ***	0.530 ***
			(0.081)	(0.082)
Agricultural income			0.000 ***	0.000 ***
			(0.000)	(0.000)
Specialized associations Participation			0.734 ***	0.982 ***
			(0.108)	(0.112)
Village terrain				−0.588 ***
				(0.042)
Geographic location				0.005 ***
				(0.002)
Promotion of agricultural machinery				0.017 ***
				(0.005)
Village cooperative				−0.340 ***
				(0.071)
Food-producing region				−0.194 ***
				(0.040)
Intercept 1	0.133	0.505 **	0.540 **	−1.380 ***
	(0.215)	(0.250)	(0.252)	(0.300)
Intercept 2	2.381 ***	2.842 ***	2.956 ***	1.181 ***
	(0.219)	(0.254)	(0.257)	(0.299)
Intercept 3	3.963 ***	4.496 ***	4.760 ***	3.034 ***
	(0.226)	(0.262)	(0.266)	(0.306)
LR_chi2	189.1	404.7	688.8	974
Prob > chi2	0	0	0	0
Pseudo R2	0.020	0.044	0.074	0.105

Standard errors in parentheses *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

5.3. Moderating Effect Analysis

To explore the differences in the role of digital literacy among farmers in the influence of labor force characteristics on DASS, this paper constructs a progressive regression model to systematically test the direct main effect of digital literacy and its moderating effect (Table 6). In Models 1–4 of Table 6, the roles of feminization, aging, and part-time work alone and all three together with digital literacy on DASS are presented, and it can be seen that digital literacy all exhibits a significant strong positive main effect, and in Model 4, for

example, controlling for the constancy of the other variables, the likelihood that a farmer's access to a higher DASS will increase by 0.1 unit for every 0.1 unit increase in digital literacy, which will increase by about 8.7%, indicating that the increase in digital literacy has a strong driving effect on DASS by farm households, and Hypothesis 2 is verified. At the same time, the three labor force characteristics of feminization, aging, and part-time work all have a positive effect on the level of service acquisition, indicating that the higher the proportion of the corresponding labor force in the farm household, the higher the likelihood of acquiring socialized services, which also verifies hypothesis 1 once again.

Table 6. Results of moderating effects.

Variable	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8
Female	0.040			0.428 **	0.433 **			0.649 **
	(0.152)			(0.201)	(0.186)			(0.256)
Old		0.204 **		0.415 ***		−0.062		0.218
		(0.096)		(0.128)		(0.110)		(0.152)
Div			0.162 *	0.191 **			0.021	0.026
			(0.087)	(0.088)			(0.105)	(0.106)
Digital literacy	0.852 ***	0.829 ***	0.856 ***	0.829 ***	1.769 ***	−0.044	0.606 ***	0.115
	(0.077)	(0.078)	(0.077)	(0.078)	(0.262)	(0.194)	(0.129)	(0.530)
Female × DL					−1.280 ***			−0.539
					(0.350)			(0.463)
Old × DL						1.163 ***		1.044 ***
						(0.237)		(0.311)
Div × DL							0.451 **	0.568 ***
							(0.188)	(0.189)
Controls	Controlled	Controlled	Controlled	Controlled	Controlled	Controlled	Controlled	Controlled
Intercept 1	−2.179 ***	−2.112 ***	−2.081 ***	−1.494 ***	−1.878 ***	−2.228 ***	−2.185 ***	−1.523 ***
	(0.236)	(0.204)	(0.211)	(0.302)	(0.250)	(0.206)	(0.215)	(0.340)
Intercept 2	0.418 *	0.487 **	0.517 **	1.111 ***	0.722 ***	0.378 *	0.412 *	1.087 ***
	(0.232)	(0.200)	(0.207)	(0.301)	(0.247)	(0.201)	(0.211)	(0.340)
Intercept 3	2.339 ***	2.410 ***	2.439 ***	3.037 ***	2.651 ***	2.318 ***	2.341 ***	3.042 ***
	(0.240)	(0.209)	(0.216)	(0.308)	(0.255)	(0.210)	(0.220)	(0.346)
Pseudo R2	0.116	0.116	0.116	0.117	0.117	0.119	0.117	0.121

Standard errors in parentheses *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Models 5–8 can be found to have heterogeneity in the moderating effect of the impact of digital literacy on labor force characteristics after the further introduction of the interaction term. In Model 5, the interaction term between feminization and digital literacy is significantly negative, suggesting that digital literacy undermines the positive push of feminization on service access to some extent. When controlling for other labor force characteristics (Model 8), this interaction effect did not reach statistical significance but still showed a negative trend. This phenomenon may be because increased digital literacy has enabled female farmers to enhance their autonomous production capacity through online learning, digital farming tools and e-commerce platforms, reducing their reliance on traditional integrated agricultural services and focusing instead on precise information support and digital technology applications, thus weakening the original positive impact of feminized characteristics on integrated service access and creating a negative moderating effect of the interaction term. Model 7 shows that the interaction term between part-time employment and digital literacy is significantly positive, suggesting that the diffusion of digital technology has made it easier for farmers engaged in off-farm employment to access agricultural socialization services through information platforms. The positive moderating effect remains robust after controlling for other labor force characteristics in Model 8, suggesting that part-time farm households can further stimulate their demand for professional services with the help of digital platforms.

Overall, digital literacy not only promotes DASS but also plays a differential role in moderating the effects of different labor force characteristics on DASS: the positive

effect of aging and part-time labor force is strengthened, while the effect of feminization is somewhat weakened. Thus, hypothesis 3 can be verified.

5.4. Heterogeneity Analysis

In the heterogeneity analysis of topography and food-producing areas (as shown in Tables A1 and A2), the regression results of Models 1 to 12 reveal the differences in the effects of labor force characteristics and digital literacy on DASS under different topographies and food-producing areas. First, labor force characteristics show positive and significant effects on DASS in the plains region, and the degree of aging has a positive effect on access to services in the main production area, while it is not significant in the rest of the region, which may be related to the differences in regional resource endowment and the strength of policy support. Surprisingly, the degree of part-time employment in the main marketing area has a negative and significant effect on service access, probably due to the lower dependence on agriculture in the main marketing area and thus less demand for agricultural socialization services.

Further moderating effect analysis revealed that the moderating effect of digital literacy on the relationship between labor force characteristics and access to agricultural socialization services varied significantly across topography and food functional areas. The interaction term between feminization and digital literacy is negatively significant in hilly areas, which is consistent with the previous findings. This phenomenon is also evident in the main marketing areas, which may be related to the strong market orientation and diversified service system in these areas, where female farmers are more inclined to utilize digital means to directly connect to the market and reduce their dependence on traditional services. Meanwhile, in the plains, the interaction term between aging and digital literacy is positively significant, consistent with the previous section, suggesting that the popularization of digital technology in the plains has lowered the threshold of information acquisition and service access, enabling older farmers to access agricultural socialization services more conveniently.

5.5. Robustness Test

In order to avoid the contingency of the impact of digital literacy and labor force characteristics on DASS, this paper uses the method of replacing the estimated model and core explanatory variables to conduct the robustness test. The "share of female labor force in household adults" and "share of non-agricultural labor force in household labor force" are used to replace feminization and part-time employment, and at the same time, the ordered probit model is chosen to replace the ordered logit model to re-estimate the data. The results (as shown in Tables A3 and A4) show that the regression results are consistent with the above, indicating that the findings of the study have a certain degree of credibility.

6. Discussion

At the same time, this study shares similarities with related research but also has specific differences. The mean value of DASS in this study was 1.163. Considering that the range of this variable is 0 to 3, it indicates that most farmers have access to fewer than three services and remain in the "basic access" stage, the proportion of female labor remains relatively stable, but there are significant differences in intergenerational structures across households, which may directly influence farmers' dependence on external services and their preferences for service needs, which is consistent with the level of service access reported in existing studies [11,37,38]. From the perspective of labor force characteristics [16]. Additionally, there is a clear differentiation between non-agricultural employment and agricultural operations among farmers, a characteristic that aligns closely with the recent

trend toward diversified rural livelihoods [14]. Farmers' digital skills exhibit a polarized distribution, indicating that the digital divide remains widespread in rural areas [42,45,46], potentially exacerbating inequalities in service access among different groups.

Empirical results show that as the feminization of agricultural labor, age structure, and occupational differentiation increase, farmers' dependence on external specialized services significantly strengthens. This finding aligns with previous research [45,47], suggesting that structural changes in the labor force will reshape the supply-demand dynamics of agricultural socialized services. Existing studies indicate that the elderly have lower internet usage rates and weaker information access capabilities, and digital literacy has not significantly influenced the elderly farmers' access to socialized services [42]. In contrast to the above conclusions, the moderation effect heterogeneity analysis in this study found that in plains and major production areas, the interaction term between aging and digital literacy was significantly positive, indicating that high levels of digital literacy can effectively compensate for the elderly farmers' deficiencies in information acquisition and technological adaptation, thereby enhancing their ability to access agricultural socialized services. This difference may be related to the well-developed infrastructure and high level of digital technology promotion in these regions, enabling even older groups to more conveniently receive training, use equipment, and convert this into service behavior. Additionally, in mountainous regions, the interaction term between part-time farming and digital literacy is significantly positive, indicating that digital technology improves mountainous farmers' information access efficiency and enhances their cognitive and acceptance capabilities regarding services, consistent with the conclusions of Gui et al. [46].

The marginal contributions of this study are mainly reflected in three aspects: firstly, it constructs a framework for analyzing the influence mechanism of "labor force characteristics-agricultural socialized service behaviors," which deepens the understanding of the behavioral economics of agricultural households in the context of agricultural labor force structural transformation; Secondly, the introduction of digital literacy expands the research dimension of explaining the adoption behavior of agricultural services from the perspective of digital empowerment; thirdly, empirical tests are carried out through micro-data, revealing the interaction mechanism between labor force characteristics and digital literacy and its regional heterogeneity, which provides theoretical support and practical references for the relevant policy formulation.

This study has some limitations. First, this study only examined the impact of labor force characteristics and digital literacy on the overall level of access to agricultural socialized services. However, agricultural socialized services encompass three stages: pre-production, production, and post-production, and different crops require different types of agricultural socialized services. Therefore, the mechanisms through which labor force characteristics and digital literacy influence access to agricultural socialized services may vary across these stages. Future research could distinguish these mechanisms to comprehensively explore the impact of labor force characteristics and digital literacy on the adoption of access to agricultural socialized services. Additionally, this study focuses on rural China, utilizing empirical analysis of household production and operation data from seven provinces in 2021. The seven provinces cover diverse terrains and grain-producing regions, ensuring natural and economic representativeness. However, China is a vast country with a large population and diverse natural conditions, and there are differences in economic and social environments across regions. The use of cross-sectional data is insufficient to effectively analyze the impacts of changes in labor force characteristics and improvements in digital literacy. Future research could appropriately expand the sample size and utilize nationwide panel data to draw more reliable conclusions.

7. Conclusions and Policy Implications

7.1. Conclusions

This paper focuses on the impact of digital literacy and labor force characteristics on DASS and empirically analyzes it using microdata from 3811 farm households in seven provinces. It is found that firstly, the feminization, aging and part-time characteristics of the labor force will have a significant impact on DASS of farm households; the higher the degree of feminization, aging and part-time, the more prominent the problem of labor force rigidity constraints, and the stronger the demand for agricultural socialized services. Secondly, digital literacy as a whole strengthens the facilitating effect of labor force characteristics on DASS, but its moderating effect shows heterogeneity. In the plains and the main food-producing areas, digital technology significantly facilitates older farmers' reliance on service adoption; in the hills and the main marketing areas, there is a dilution of the positive effect of digital literacy on feminization due to the solidification of gender roles and the complexity of the market. In addition, the synergistic effect of part-time employment and digital literacy is particularly prominent in topographically complex regions, suggesting that digital technology can break through geographic constraints and activate the latent service demand of non-farm labor.

7.2. Policy Implications

Based on the above findings, policymakers should formulate locally appropriate strategies for the extension of agricultural socialized services according to the actual situation in different regions, especially in terms of labor force structure and the degree of digital technology penetration, by adopting targeted measures. As a result, this paper proposes the following insights:

(1) Strengthen the popularization and publicity of agricultural socialized services and formulate differentiated socialized service extension strategies. For areas with a high proportion of female or serious aging and remote mountainous areas, promote mechanized services that are highly adaptable and easy to operate and tailored to the schedules and skill requirements of women and older people, and at the same time, strengthen training for farmers on how to choose the right services; for areas with large-scale agriculture such as the plains, promote the specialization and industrialization of socialized services, encourage agricultural enterprises to provide farmers with more personalized services, and promote the intensification and modernization of agricultural operations.

(2) Strengthen the digital technology training of rural laborers through various ways, and design differentiated digital literacy improvement programs by taking into account differences in laborer characteristics. At the same time, in cooperation with local agricultural cooperatives, rural schools, agricultural technology extension agencies, and other organizations, regular online and offline training courses should be organized. These courses should be tailored to the specific characteristics of different farmer groups. For part-time farmers, flexible training modules should be developed to enable them to conveniently access digital knowledge and service information even when time is limited. Training content should cover how to use smartphones, how to access agricultural information platforms, how to obtain online agricultural machine leasing and maintenance services, and how to use the Internet to make decisions on agricultural production. By improving the digital literacy of farmers and lowering their cognitive threshold of socialized services, farmers are encouraged to actively adopt advanced agricultural socialized services and improve agricultural production efficiency.

(3) Deepen the digitalization of agricultural services and promote the deep integration of digital technology and agricultural production. Referring to Zhejiang Province's "Zhejiang Agricultural Service" platform model, a unified information platform for agricultural socialized services should be constructed, integrating agricultural technology, agricultural machinery services, agricultural finance, climate information, and other resources to provide one-stop services, making it easy for farmers to obtain services simply and quickly. Each region should design an appropriate digital platform according to its own geographic, economic and cultural characteristics to ensure that the platform can meet the needs of different groups of farmers, especially in remote areas, and should increase the construction of Internet infrastructure, such as 5G base stations and fiber-optic network coverage, to ensure that farmers can access and use the platform smoothly and to reduce the cost of searching for services for farmers.

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Abbreviations

The following abbreviations are used in this manuscript:

DASS The degree of adoption of agricultural socialized services

Appendix A

Table A1. Topographic heterogeneity and regional variability in baseline regressions.

Variable	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
	Plains	Hills	Mountains	Main Production	Main Consumption	Balanced Area
Female	0.598 **	0.085	0.634	0.385	−0.127	0.599
	(0.298)	(0.387)	(0.399)	(0.295)	(0.411)	(0.398)
Old	0.957 ***	0.176	0.220	0.341 *	−0.028	0.296
	(0.194)	(0.238)	(0.256)	(0.175)	(0.306)	(0.268)
Div	0.352 **	−0.156	−0.020	−0.020	0.170	−0.368 **
	(0.138)	(0.164)	(0.184)	(0.123)	(0.220)	(0.179)
Controls	Controlled	Controlled	Controlled	Controlled	Controlled	Controlled
Intercept 1	−0.679	−1.139 **	0.063	−1.823 ***	−1.686 **	−0.095
	(0.448)	(0.550)	(0.613)	(0.415)	(0.674)	(0.522)
Intercept 2	2.406 ***	1.434 ***	2.294 ***	0.786 *	2.078 ***	2.526 ***
	(0.450)	(0.552)	(0.618)	(0.412)	(0.655)	(0.530)
Intercept 3	4.061 ***	4.072 ***	4.291 ***	3.212 ***	3.808 ***	4.741 ***
	(0.459)	(0.575)	(0.645)	(0.427)	(0.667)	(0.579)
Pseudo R2	0.115	0.0842	0.0855	0.0613	0.193	0.0510

Standard errors in parentheses *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table A2. Topographic heterogeneity and regional variability in moderating effects.

Variable	Model 7	Model 8	Model 9	Model 10	Model 11	Model 12
	Plains	Hills	Mountains	Main Production	Main Consumption	Balanced Area
Female	0.314	0.975 *	0.865 *	−0.006	0.461	1.434 ***
	(0.401)	(0.498)	(0.500)	(0.371)	(0.599)	(0.490)
Old	0.186	0.434	0.141	−0.145	−0.114	0.516
	(0.243)	(0.291)	(0.301)	(0.210)	(0.385)	(0.316)
Div	0.563 ***	−0.120	−0.333	−0.025	0.623 **	−0.517 **
	(0.176)	(0.205)	(0.218)	(0.152)	(0.292)	(0.218)
Digital literacy	1.267	2.969 ***	−0.729	−1.150	4.366 ***	2.910 **
	(0.790)	(1.026)	(1.109)	(0.782)	(1.151)	(1.187)
Female × DL	−0.274	−2.521 ***	−0.670	1.079	−1.315	−3.084 ***
	(0.662)	(0.909)	(0.977)	(0.683)	(0.936)	(1.052)
Old × DL	1.255 **	−0.985 *	0.550	1.294 ***	0.312	−0.930
	(0.488)	(0.583)	(0.641)	(0.434)	(0.722)	(0.713)
Div × DL	−0.123	−0.099	1.144 ***	0.143	−1.367 ***	0.441
	(0.299)	(0.351)	(0.404)	(0.269)	(0.495)	(0.426)
Controls	Controlled	Controlled	Controlled	Controlled	Controlled	Controlled
Intercept 1	−1.139 **	−0.192	0.079	−2.354 ***	−0.588	0.559
	(0.530)	(0.644)	(0.686)	(0.478)	(0.813)	(0.593)
Intercept 2	2.202 ***	2.409 ***	2.328 ***	0.301	3.548 ***	3.202 ***
	(0.533)	(0.649)	(0.691)	(0.475)	(0.821)	(0.603)
Intercept 3	4.161 ***	5.055 ***	4.343 ***	2.785 ***	5.763 ***	5.428 ***
	(0.542)	(0.670)	(0.715)	(0.486)	(0.840)	(0.647)
Pseudo R2	0.181	0.0898	0.0908	0.0727	0.297	0.0566

Standard errors in parentheses *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table A3. Results of the robustness test analysis of the benchmark regression.

Variable	Replacement Estimation Model				Substitution of Explanatory Variables			
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8
Female	0.241 **	0.256 **	0.287 **	0.279 **	0.319 *	0.377 *	0.346 *	0.330 *
	(0.113)	(0.114)	(0.115)	(0.116)	(0.193)	(0.193)	(0.195)	(0.198)
Old	0.648 ***	0.475 ***	0.353 ***	0.328 ***	1.069 ***	0.818 ***	0.571 ***	0.529 ***
	(0.070)	(0.071)	(0.072)	(0.073)	(0.130)	(0.133)	(0.136)	(0.136)
Div	0.466 ***	0.421 ***	0.243 ***	0.132 ***	0.773 ***	0.718 ***	0.393 ***	0.174 **
	(0.046)	(0.047)	(0.049)	(0.051)	(0.079)	(0.080)	(0.085)	(0.088)
Controls		Controlled	Controlled	Controlled		Controlled	Controlled	Controlled
Intercept 1	0.058	0.267 *	0.282 *	−0.830 ***	0.019	0.543 **	0.463 *	−1.524 ***
	(0.125)	(0.144)	(0.145)	(0.173)	(0.213)	(0.253)	(0.255)	(0.299)
Intercept 2	1.427 ***	1.678 ***	1.732 ***	0.691 ***	2.266 ***	2.882 ***	2.878 ***	1.035 ***
	(0.126)	(0.146)	(0.147)	(0.172)	(0.216)	(0.258)	(0.260)	(0.297)
Intercept 3	2.310 ***	2.598 ***	2.719 ***	1.717 ***	3.848 ***	4.538 ***	4.682 ***	2.887 ***
	(0.129)	(0.149)	(0.151)	(0.175)	(0.223)	(0.265)	(0.269)	(0.304)
Pseudo R2	0.0212	0.0435	0.0725	0.105	0.0202	0.0441	0.0743	0.105

Standard errors in parentheses *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table A4. Results of robustness test analysis of moderating effects.

Variable	Replacement Estimation Model				Substitution of Explanatory Variables			
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8
Female	0.277 **			0.393 ***	0.408 **			0.658 **
	(0.109)			(0.149)	(0.186)			(0.256)
Old		−0.049		0.124		−0.031		0.256 *
		(0.064)		(0.089)		(0.111)		(0.153)
Div			0.032	0.033			0.023	0.033
			(0.062)	(0.062)			(0.105)	(0.106)
Digital literacy	1.019 ***	−0.024	0.326 ***	0.086	1.760 ***	−0.056	0.605 ***	0.076
	(0.151)	(0.107)	(0.073)	(0.302)	(0.262)	(0.194)	(0.130)	(0.530)
Female × DL	−0.765 ***			−0.339	−1.269 ***			−0.519
	(0.200)			(0.266)	(0.350)			(0.463)
Old × DL		0.647 ***		0.574 ***		1.174 ***		1.068 ***
		(0.132)		(0.174)		(0.237)		(0.311)
Div × DL			0.261 **	0.338 ***			0.454 **	0.574 ***
			(0.108)	(0.109)			(0.188)	(0.189)

Table A4. *Cont.*

Variable	Replacement Estimation Model				Substitution of Explanatory Variables			
	Model 1	**Model 2**	**Model 3**	**Model 4**	**Model 5**	**Model 6**	**Model 7**	**Model 8**
Controls	Controlled	Controlled	Controlled	Controlled	Controlled	Controlled	Controlled	Controlled
Intercept 1	−1.092 *** (0.144)	−1.327 *** (0.119)	−1.277 *** (0.124)	−0.894 *** (0.197)	−1.852 *** (0.250)	−2.154 *** (0.208)	−2.134 *** (0.217)	−1.428 *** (0.342)
Intercept 2	0.442 *** (0.143)	0.210 * (0.117)	0.256 ** (0.122)	0.648 *** (0.197)	0.752 *** (0.247)	0.456 ** (0.204)	0.467 ** (0.213)	1.188 *** (0.342)
Intercept 3	1.511 *** (0.146)	1.284 *** (0.120)	1.325 *** (0.125)	1.728 *** (0.199)	2.681 *** (0.256)	2.396 *** (0.212)	2.396 *** (0.221)	3.143 *** (0.348)
Pseudo R2	0.116	0.118	0.116	0.120	0.118	0.120	0.118	0.122

Standard errors in parentheses *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

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